Question 11
The PoliMi data center has two simultaneously active (redundant) cooling distribution systems. Knowing that the availability of the cooling system should be 99.96% and the MTTR of a single distribution system is 15 days, what is the minimum MTTF needed to reach the required availability value? Use at least 4 decimal digits for all the intermediate calculations.

$$A_T = 1 - (1 - A_i)^2 > 0{,}9996 \quad \text{where} \quad A_i = \frac{MTTF}{MTTF + MTTR}$$

$$\left(1 - \frac{MTTF}{MTTF + 15}\right)^2 < 0{,}0004 \rightarrow \frac{MTTF}{MTTF + 15} > 1 - \sqrt{0{,}0004}$$

$$\downarrow$$

$$MTTF\sqrt{0{,}0004} > 15\left(1 - \sqrt{0{,}0004}\right)$$

$$MTTF > \frac{15\left(1 - \sqrt{0{,}0004}\right)}{\sqrt{0{,}0004}} = \boxed{735 \text{ days}}$$

Question 12
Consider a RAID 4 configuration composed of an array with 10 disks. What is the minimum number of I/O operations requested to update one block of a single data disk (considering the sum over the entire set of disks)?

$\boxed{\text{4 I/O operations}}$:
1 read old data
1 read old parity
- calculate new parity (NO I/O)
1 write new data
1 write new parity

Question 13
A company wants to evaluate the performance of the services provided to its users. The computer system includes two servers Sv_1 and Sv_2. The system is initially considered as an open queue network model and the following measurements were obtained during 10-minute monitoring:

- Number of requests served at system level: $C = 300$
- Number of requests served by Sv_1: $C_{Sv1} = 600$
- Number of requests served by Sv_2: $C_{Sv2} = 100$
- Utilization $U_{Sv1} = 0.3333$
- Utilization $U_{Sv2} = 0.250$

What is the busy time B_{Sv1} and B_{Sv2}?

$$X_k = \frac{C_k}{T} \qquad U_k = \frac{B_k}{T} \qquad S_k = \frac{B_k}{C_k}$$

$$\rightarrow B_k = U_k \cdot T$$

$$B_1 = U_1 \cdot T = 3{,}333 \text{ min} = \boxed{200 \text{ s}}$$

$$B_2 = U_2 \cdot T = 2{,}5 \text{ min} = \boxed{150 \text{ s}}$$

Question 14
Considering the system described in the previous question, what is the service demand for the two servers (D_{Sv1} and D_{Sv2})?

$$X_k = V_k X \qquad D_k = S_k V_k$$

$$U_k = S_k X_k = S_k V_k X = D_k X = D_k \frac{C}{T} \rightarrow D_k = \frac{U_k T}{C}$$

$$\boxed{D_1 = 0{,}666 \text{ sec}}$$

$$\boxed{D_2 = 0{,}5 \text{ sec}}$$

Question 15
Considering now the same system as a closed model with a think time $Z = 10$ sec, and the same values for the demand D_{Sv1} and D_{Sv2} calculated for the open model. In the context of the asymptotic bounds, what is the system throughput upper bound and response time lower bound for N= 40 users?

$$\max(D,\ ND_{MAX} - Z) \leq R(N) \leq ND$$

$$\max(1{,}1666,\ 40 \cdot 0{,}\bar{6} - 10)$$

$$\boxed{16{,}\bar{6}\ \text{sec}}$$

$$\frac{N}{ND+Z} \leq X(N) \leq \min\left(\frac{N}{D+Z},\ \frac{1}{D_{MAX}}\right)$$

$$\min\left(\frac{40}{1{,}1\bar{6}+10},\ \frac{1}{0{,}\bar{6}}\right)$$

$$\boxed{1{,}5\ \frac{req}{s}}$$

Question 16
In the same system above, if you consider two instances of Sv_1, how do the bounds change?

$$D_k = \frac{U_k T}{2C} \longrightarrow D_1 = 0{,}\bar{3}\ \text{sec}\ ,\quad D_2 = 0{,}5\ \text{sec}$$

$$\max(D,\ ND_{MAX} - Z) \leq R(N) \leq ND$$

$$\max(0{,}8333,\ 40 \cdot 0{,}5 - 10)$$

$$\boxed{10\ \text{sec}}$$

$$\frac{N}{ND+Z} \leq X(N) \leq \min\left(\frac{N}{D+Z},\ \frac{1}{D_{MAX}}\right)$$

$$\min\left(\frac{40}{0{,}8\bar{3}+10},\ \frac{1}{0{,}5}\right)$$

$$\boxed{2\ \frac{req}{s}}$$